

PRAMANEX

LEDGER-Q

QUALITY, COMPLIANCE & AI ASSURANCE OS

Enterprise SaaS Master Blueprint - joined quality system + AI trust control, workflow, website, data, agents, tests and Gemini build specification

Dual operating logic Quality stream: source -> requirement -> gap/event -> investigation -> risk -> RCA/CAPA/change -> training/action -> human QA decision -> effectiveness -> inspection evidence. AI stream: model/use -> provenance -> evaluation -> grounding -> bias/consistency -> risk -> human override -> release status -> monitor/change/retest. Both converge at named human authority and reconstructable evidence.

SOURCE	REQUIREMENT	EVENT/GAP	INVESTIGATE	RISK	CAPA/CHANGE	HUMAN QA	EFFECTIVENESS	AUDIT
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AI USE	PROVENANCE	EVALUATE	GROUND	CHALLENGE	HUMAN OVERRIDE	RELEASE	MONITOR	RETEST
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Truth boundary: 20 features are a controlled product design target unless runnable artifacts and executed evidence prove implementation. AI never autonomously determines regulated root cause, disposition, closure or release.

0. Product identity and architecture lock

Element	Status	Build rule
F01-F20	Canonical integrated feature set	Preserve current IDs; quality/compliance + AI assurance remain one product.
L01-L12	Canonical shared layers	All 20 features reuse the same implementation/control layers.
LQ-P01-LQ-P05	Proposed differentiator controls	Child controls only; not verified world-first claims.
LQ-R01-LQ-R04	Proposed research/challenge controls	Use for adverse evaluation and evidence quality; retain failures.
Standalone GxP Training AI OS	Future extension	Do not build as a separate module now. F10 basic training impact/retraining traceability remains part of quality workflow.
Website / SaaS / agents	Proposed implementation design	Must preserve evidence, versioning, human control and release-state semantics.

1. Canonical 20 joined features

ID	Feature	Primary work	Layers
F01	Controlled Quality Source & Document Registry	Approved SOPs/policies/forms/quality sources with owner, permission, effective date, hash and version history.	L01-L04, L06
F02	Requirement Extraction & Exact Evidence Traceability	Structured requirements with exact source/version/location link.	L04-L07, L09-L11
F03	Regulation-to-SOP / Requirement Mapping	Connect external requirements to applicable SOPs/procedures/processes/versions with rationale.	L04-L07, L09-L11
F04	Compliance Gap, Conflict & Evidence-Sufficiency Control	Detect missing/conflicting/stale/unsupported evidence; maintain assurance debt and block unsafe progression.	L05-L08, L10-L11
F05	Quality Risk, Severity & Confidence Assessment	Separate impact/severity, likelihood, controls, evidence sufficiency and confidence.	L06, L08, L10-L11
F06	Deviation Investigation Review	Reconstruct chronology/evidence/gaps; AI may propose hypotheses but never determine root cause/disposition.	L04-L09, L10-L11
F07	OOS/OOT Investigation Review	Reconstruct laboratory event evidence/conflicts/missing data/human disposition and any AI contribution.	L04-L09, L10-L11
F08	Bidirectional Change-Control & Impact Assessment	Quality change can stale dependent AI assurance; AI/model/prompt/tool/data/policy change can reopen affected SOP/training/quality review.	L04, L06-L12
F09	RCA-CAPA Lifecycle & Effectiveness Verification	Evidence-linked hypothesis -> CAPA candidate -> owner/action -> implementation -> effectiveness -> human closure.	L04, L06-L12
F10	Training Impact & Retraining Traceability	Identify affected roles/training after SOP/CAPA/system change and preserve completion against correct version.	L02, L04, L06-L08, L10-L12

ID	Feature	Primary work	Layers
F11	Audit / Inspection Finding & Readiness Management	Finding -> gap -> action/CAPA -> owner -> response -> closure with reconstructable inspection evidence and open debt.	L02, L04, L06-L08, L10-L12
F12	PQR/APQR & Quality Trend Review	Aggregate event/CAPA/assurance patterns into human-reviewed trends; correlation never becomes automatic conclusion.	L04-L08, L11-L12
F13	Owner, Deadline, Escalation & Closure Control	Assign accountability, dates, escalation and evidence-backed closure for quality/revalidation/reopened controls.	L02, L08, L10-L12
F14	Human QA Review, Override & Controlled Correction	Named reviewer approves/revises/requests evidence/rejects/suspends/escalates; corrections cannot silently self-train/change policy.	L02, L04, L08, L10-L11
F15	Audit/Version Reconstruction, Event Forensics & Controlled Export	Reconstruct source -> analysis -> AI run -> human use/correction -> final decision; issue version-bound evidence/release passport.	L02-L04, L07-L12
F16	AI Model / Use / Context & Release Registry	Register model/version/owner/intended use/exclusions/dependencies/reviewer and VALID/SUSPENDED/EXPIRED/REVOKED/REVALIDATED state.	L01-L04, L06-L08, L10-L11
F17	AI Provenance, Transparency & Contribution Record	Preserve model, prompt, tools, configuration, inputs, retrieval evidence, output and human interaction per AI-assisted run.	L03-L07, L09-L11
F18	AI Evaluation, Grounding & Hallucination Suite	Frozen expected-vs-actual tests, citation/evidence checks and unsupported-output detection; assessor conflicts visible.	L05-L07, L09-L11
F19	Bias, Edge-Case & Consistency Verification	Defined subgroup/edge/adverse cases, repeated-run variation and model/version comparison with denominators/limits.	L04, L06, L09-L11
F20	Assurance Monitoring, Regression, Blast Radius & Revalidation	Monitor failures/abstention/task success, replay failures, calculate debt/blast radius, invalidate stale assurance and trigger retest/release-state update.	L04, L06-L12

2. Final 12 shared architectural layers

Layer	Name	Work inside LEDGER-Q
L01	Workspace & Tenant Isolation	Separate organizations and prevent cross-tenant quality/model/evidence leakage.
L02	Identity, Roles & Approval	Enforce QA reviewer, owner, SME, administrator and approver authority.
L03	Authorized Source & Intake	Admit only permitted SOPs, records, datasets, model inputs and approved evidence.
L04	Storage, Metadata, Versioning & Release Records	Preserve originals, hashes, model/prompt/config versions, before/after lineage, AI runs and release passports.
L05	Domain Parsing & Normalization	Parse documents/tables/quality records into typed records without losing coordinates.
L06	Source / Policy / Context & Staleness Governance	Apply source authority, policy, intended use, context-of-use, staleness and bidirectional invalidation.
L07	Evidence Graph, Retrieval, Assurance Debt & Blast Radius	Connect requirements, SOPs, events, CAPA, training, product/batch, AI, decisions and dependent outputs.

Source-derived LEDGER-Q core + clearly marked proposed enterprise implementation

Layer	Name	Work inside LEDGER-Q
L08	Workflow State, Reopening & Orchestration	Control queues/waits/retries/escalation/corrections/reopened reviews/revalidation and safe failures.
L09	Multi-Agent & Multi-Model Routing	Route bounded requirement, deviation, OOS/OOT, change, CAPA and AI-assurance specialist tasks.
L10	Safety, Security & Human Decision	No autonomous regulated disposition/root cause/closure; enforce override, suspension and controlled release.
L11	Evaluation, Audit, Correction Firewall & Revalidation	Run fixed/adverse tests, grounding, bias/consistency, regression/retest, audit/KPIs; human correction never silently rewrites memory/policy.
L12	Integration, Operations & Scale	APIs, QMS/RIM boundaries, monitoring, notifications, exports, recovery/rollback and later scale.

3. Proposed LEDGER-Q controls - child enhancements

ID	Control	Behavior	Proof
LQ-P01	Cross-domain dependency invalidation	A quality/SOP/process/data/training change stales dependent AI assurance; an AI/model/prompt/tool/data change reopens affected SOP/validation/training/quality review.	Test both directions without rewriting history.
LQ-P02	Reconstructable failure / AI-contribution forensics	For deviation/OOS/CAPA or AI-evidence failure, reconstruct exact AI run, evidence, output and human use; first failure remains preserved.	Return contribution candidate or UNKNOWN, never autonomous root cause.
LQ-P03	Human-correction regression firewall	A human correction may become a proposed regression case only after QA approval/versioning; never silently train or alter benchmark/memory.	Correct v1 -> approve regression candidate -> version -> retest.
LQ-P04	Assurance debt + conflicting-assessor / blast-radius panel	Show stale/missing/conflicting assurance, deterministic/model/human disagreement and exact affected objects. Missing evidence cannot be outvoted.	UNKNOWN never scored as safe; affected dependencies visible.
LQ-P05	Non-gaming trust report	Show task success, abstention, critical failure cost, review burden and denominators separately.	A system that refuses everything cannot look perfect.

Research / challenge controls

ID	Challenge
LQ-R01	Pharma critical-error challenge pack: units, operators, batch, version and evidence mutations.
LQ-R02	Evidence-removal challenge: remove support and require abstention or valid alternate source.
LQ-R03	Task-specific model-change comparison: critical paired cases cannot be hidden by aggregate score.
LQ-R04	Failure-location report: identify first evidenced break; unresolved cause stays unresolved.

2A. Workspace-scoped ingestion + bounded-routing enhancements

These controls preserve the canonical layer names. L01 continues to own tenant/workspace isolation; authorized intake remains governed by L03; the multimodal parsing execution is performed under L05. L09 gains an explicit outage/fallback protocol.

L01/L03/L05 capability - Vision-LLM & Tabular Extraction Pipeline

Input	Pipeline behavior	Safety / evidence rule
OOS/OOT lab evidence	Route chromatograms, HPLC graphs, handwritten lab notes and complex nested tables to a Vision-LLM / document-AI path when simple parsing is insufficient.	Tenant/source permissions checked before processing; original artefact is retained.
Structured output	Convert observed content into typed JSON with source version, page/region coordinates, table/series identifiers, raw observation, units, method, confidence and parser/model identity.	JSON is a candidate representation; ambiguous or unreadable content becomes UNKNOWN / NEEDS_REVIEW.
Dual-path extraction	Use deterministic text/table extraction first where reliable; vision route is conditional for complex visual evidence.	Do not replace source chromatogram/graph/note with generated text. Human review remains required for quality disposition.

Data fidelity rule Target a high-fidelity/loss-minimizing JSON representation while preserving original files, image regions and coordinates. Do not claim "lossless" extraction unless validated against representative laboratory fixtures.

L09 protocol - Deterministic Fallback & Dead-Letter Queue (DLQ)

Level	Protocol	LEDGER-Q behavior
1 - Retry	3 automatic retries with exponential backoff for eligible AI/API failures.	Idempotent retries; failure remains visible and traceable.
2 - Queue	Non-critical asynchronous AI summaries and similar tasks go to a DLQ during provider downtime.	Persist exact revision, run ID, failure reason and retry metadata; auto-process after recovery under current authorization/policy.
3 - Deterministic bypass	Critical GxP/quality operations switch to Rule-Based / Manual Mode.	Quality events, required evidence, deterministic rules, owner/action routing and named human QA work remain available even when AI is unavailable.

Continuity rule Factory/quality workflow must not depend on LLM availability for required critical operations. AI outage cannot auto-close, auto-release or suppress a quality state.

4. Simultaneous feature-layer linkage

Stage	Features	Layers	Outcome
1. Control truth	F01 -> F02 -> F03 -> F04	L01-L07	Approved source -> exact requirement -> mapped control -> gap/assurance debt.
2. Understand quality risk	F05 + F06 + F07 + F08	L04-L11	Risk/investigation + bidirectional impact/dependency reopening.
3. Correct & close	F09 + F10 + F13	L02, L04, L06-L12	RCA/CAPA -> training/action -> owner -> effectiveness -> closure.
4. Oversight / inspection	F11 + F12 + F15	L02, L04, L06-L12	Findings, trends, event forensics, evidence/passport export.
5. Human authority	F14 governs F05-F20	L02, L08, L10-L11	Approve/revise/request evidence/reject/suspend/escalate; corrections controlled.
6. Govern AI	F16 -> F17 -> F18 -> F19 -> F20	L01-L12	Register -> provenance -> evaluate -> challenge -> monitor/revalidate -> release state.
7. Continuous trust loop	Any material quality/AI change -> F08/F20 -> F14 -> retest -> F15	L06-L12	Invalidate stale assurance, calculate blast radius, reopen controls, update passport after human approval.

5. Integrated runtime - two streams, one human boundary

Quality / compliance stream

SOURCE	REQUIREMENT	GAP/EVENT	INVESTIGATION	RISK	RCA/CAPA/CHANGE	TRAIN/ACTION	HUMAN QA	EFFECTIVENESS	AUDIT
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AI assurance stream

AI/USE	PROVENANCE	EVALUATE	GROUND	BIAS/EDGE	RISK	HUMAN OVERRIDE	RELEASE	MONITOR	RETEST
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Convergence Any material change is checked both directions through the evidence graph. Affected controls are reopened, human reviewers decide, and only approved retest can update the version-bound release/passport state.

Release-state semantics

State	Meaning
VALID	Approved evidence and context remain current.
SUSPENDED	Temporary hold while issue/change is investigated.
EXPIRED	Time/context/version trigger requires revalidation.
REVOKED	Evidence/control failure makes prior release unusable.
REVALIDATED	Required retest + human approval restore controlled use.

5A. F13/F14 Human-in-the-Loop Authority - E-Signature Gateway

Sub-feature: **E-Signature Gateway (21 CFR Part 11 compliance-targeted)**. The control binds status-changing QA decisions to an authenticated named human and exact evidence revision.

Control	Required behavior	Audit evidence
Status-transition gate	Configured transitions such as Draft -> Approved or Draft -> Rejected cannot complete until the signature gateway succeeds.	Object/revision, prior/new state, workflow/correlation ID and policy version.
2-factor authentication	Strict Password + OTP or approved biometric second factor at the signing event.	User_ID, role/authority, authentication event, UTC timestamp and factor result.
Reason for Signature	Mandatory controlled choice: Authored, Reviewed or Approved (plus bounded comment if configured).	Reason code is stored with exact decision revision.
Immutable audit log	Create real-time UTC timestamped append-oriented audit entry with cryptographic hash of the signed revision/decision metadata.	Record hash, previous/chain reference where implemented, actor, status change, reason and timestamp.
Revision integrity	A material edit after signature creates a new revision and invalidates the signed approval state for affected content.	Prior signed state remains reconstructable; new review/signature is required.

State-machine addition

DRAFT -> HUMAN_REVIEW_REQUIRED -> SIGNATURE_REQUIRED -> AUTHENTICATED -> APPROVED / REJECTED.
Authentication failure, stale revision, missing reason or hash mismatch must route to HOLD / REAUTHENTICATE / RE-REVIEW.

Part 11 truth boundary This is a compliance-targeted blueprint control, not proof that a deployed system is 21 CFR Part 11 compliant. A real compliance claim requires validated software, identity/access controls, SOPs, record-retention policies, operational governance and executed qualification/testing.

Human authority remains final The gateway authenticates and evidences the human act. AI cannot become the approver, root-cause owner, disposition authority or release authority.

6. Detailed user journeys

Journey	Human path	System responsibility
Compliance mapping	Add controlled requirement source -> extract requirement -> map to SOP/process -> review gap.	Preserve exact source/version; do not auto-close missing coverage.
Deviation	Open event -> chronology/evidence -> hypotheses/gaps -> risk -> CAPA candidate -> QA decision -> effectiveness.	AI may support hypotheses only; named human owns root cause/disposition.
OOS/OOT	Evidence timeline -> conflicting/missing lab evidence -> review -> decision/CAPA.	Preserve source methods/results and any AI contribution.
Change control	Propose quality/AI change -> compute affected SOP/process/training/model/decisions -> reopen dependent controls -> review/retest.	Bidirectional stale/invalidation graph.
Audit / inspection	Finding -> evidence gap -> owner/action/CAPA -> response -> closure -> reconstruction package.	Open debt remains visible until evidence-backed closure.
AI onboarding	Register model/use/context -> provenance -> fixed evaluation -> grounding/bias/edge tests -> QA review -> release state.	No release without intended use, evidence and human approval.
AI change / regression	Prompt/model/data/tool/policy changes -> blast radius -> mark stale -> rerun critical tests -> review -> state update.	Critical paired failures cannot be hidden by aggregate score.
Historical replay	Select any past decision/event -> reconstruct exact sources, AI run, corrections, rationale and status.	Use original versions; never regenerate history with today's state.

10. Enterprise SaaS website - experience architecture

Design objective LEDGER-Q should feel like a regulated enterprise workbench, not a marketing mock-up. Every screen should help a real user understand evidence, status, ownership, risk and the next permitted action.

Primary user roles

Role	Primary job	Default authority
QA Reviewer / Approver	Review quality events, risk, CAPA/change, evidence and release status.	Approve/revise/request evidence/reject/suspend/escalate.
Quality Investigator / Owner	Build event chronology, collect evidence, propose actions and complete effectiveness work.	Candidate work; no autonomous closure.
AI Assurance Reviewer	Register/evaluate AI use, inspect provenance, failures, bias/edge cases and revalidation.	Recommend/approve within assigned authority.
SME / Process Owner	Provide process/SOP/product context and own assigned corrective actions.	Domain evidence and action ownership.
Admin / Validation	Configure roles, evidence policies, workflows, test sets and recovery.	Administrative/validation controls; no automatic QA authority.

Authenticated product navigation

Route / page	Purpose	Main objects / actions
/app	Assurance Command Center	Open quality events, assurance debt, AI release states, overdue actions, inspection readiness.

Route / page	Purpose	Main objects / actions
/sources	Controlled Sources	SOP/policy/record/dataset/model-input registry and version health.
/requirements	Requirements & SOP Map	Extracted requirements, coverage mapping, gaps and evidence sufficiency.
/events	Quality Events	Deviation and OOS/OOT queues; chronology/evidence/risk.
/changes	Change & Blast Radius	Quality/AI change graph, affected SOP/training/product/model/decision objects.
/capa	RCA-CAPA & Effectiveness	Hypotheses, actions, owners, implementation and effectiveness evidence.
/training	Training Impact	Affected roles, required retraining/review and completion by version.
/inspection	Inspection Readiness	Findings, evidence gaps, response/closure and readiness packet.
/trends	PQR/APQR & Trends	Reviewable quality patterns and assurance exceptions with denominators.
/ai-registry	AI Use & Release Registry	Model/use/context/dependencies/reviewer/release passport.
/ai-evaluation	AI Evaluation Lab	Frozen cases, grounding, hallucination, bias/edge, model-change comparison.
/review	Human QA Review Queue	Exact revision decision + residual risk + correction firewall.
/audit	Replay / Forensics / Export	Historical reconstruction, AI contribution replay, controlled evidence/passport export.
/settings	Organisation / Controls	Roles, source policy, workflow rules, test packs, integrations, notifications.

Command center / dashboard

Card / panel	What it answers	Interaction
Open quality risk	What needs QA attention now?	Open prioritized event/review queue.
Assurance debt	What evidence is missing/stale/conflicting?	Open debt + dependency panel.
Release states	Which AI uses are valid/suspended/expired/revoked?	Open registry/passport.
Blast radius	What became stale after this change?	Open dependency graph.
CAPA effectiveness	Which actions lack effectiveness evidence?	Open CAPA workbench.
Inspection readiness	Which findings/gaps remain open?	Open inspection evidence chain.
Critical AI failures	Which frozen/edge cases failed?	Open evaluation trace and first failure.
Overdue owners	What actions/revalidation are past due?	Open owner/deadline/escalation queue.

Visual language

- Desktop-first enterprise shell: persistent left rail, top context bar, global search, workspace switcher, notification center, user/role menu.

- Color system: deep navy foundation; restrained cyan/blue for active evidence; amber for needs-review; red only for blocked/critical; green only for verified/approved states.
- Typography: modern grotesk/sans family; compact 12-14 px operational text, 16-20 px section titles, 28-40 px hero statements. Use clear hierarchy, not decorative typography.
- 3D/motion: use subtle depth in the public landing page and architecture visualization only. Inside regulated work screens, motion is functional: state transition, trace highlight, dependency pulse and review confirmation.
- Never hide uncertainty behind animation. State, source version, evidence completeness, human authority and failure reason remain textually visible.

Public landing page

Section	Content	Motion / visual
Hero	LEDGER-Q identity + one-sentence problem/outcome + controlled demo CTA.	Slow 3D evidence-network / document-stack motion; no distracting particles.
Problem	Quality systems often fragment requirements, deviations, OOS/OOT, CAPA, change, training and inspection evidence while AI-assisted work adds a second trust problem: intended use, provenance, evaluation, change and release status.	Before/after workflow strip.
How it works	Quality evidence and AI assurance share one versioned evidence graph, one accountable human boundary and one reconstructable history.	Scroll-linked stage illumination.
Evidence & human control	Exact source/version links, human gate, failure states, audit reconstruction.	Interactive evidence-chain hover.
Product screens	Command center, workbench, review, audit/export screenshots.	Subtle parallax cards.
Architecture	Frontend -> API -> orchestration -> data/evidence -> evaluation -> human gate.	Layered 3D stack, reduced-motion fallback.
Proof / limitations	Executed tests, demo scope, known limits, synthetic/public-safe data label.	Static; evidence-first.
CTA	Open controlled demo / view case study / GitHub evidence as actually available.	Minimal hover only.

11. Frontend, backend, database and infrastructure

Reference implementation stack

Plane	Recommended implementation	Boundary
Frontend	Next.js / React + TypeScript + accessible component system; private operational UI may use Streamlit for a bounded prototype.	Do not confuse a public portfolio shell with the controlled review application.
API	Python + FastAPI + Pydantic / JSON Schema.	Typed input/output; reject malformed or unauthorized work before model calls.
Auth / tenancy	Supabase Auth + PostgreSQL row-level security or equivalent.	Tenant, role and purpose checks at relational, object, retrieval and export paths.
Database	PostgreSQL + versioned relational records; pgvector only where retrieval adds value.	Relational truth first; vector retrieval never replaces source/version evidence.
Object storage	Versioned controlled documents, manifests and exported evidence packages.	Original bytes retained; safe content types and access policies.
Automation	n8n parent workflow + named bounded specialist subflows.	Deterministic routing, waits, retries and audit calls owned by one parent orchestrator.

Plane	Recommended implementation	Boundary
AI runtime	Pinned model/configuration gateway with explicit fallback and recorded model identity.	No silent model substitution; task-specific evaluation before release.
Evaluation	pytest + fixed fixtures + structured outcome logs; optional RAG evaluation where appropriate.	Report denominators, failures and limitations.
Observability	Structured logs + trace IDs + workflow metrics + product analytics.	Never send regulated/confidential payloads to analytics by default.
Deployment	Containerized API/worker; managed PostgreSQL/storage; scheduler/worker runs independently of browser.	Live means deployed endpoint actually tested; production requires additional security/recovery/validation work.

Core data objects

Object	Minimum fields / relationships
workspaces	tenant, site/product/process context, quality policy version, workflow/schema version
controlled_sources	SOP/policy/form/record/dataset IDs, owner, permission, effective date, version, hash, predecessor
requirements	typed requirement, exact source span, applicability rationale, evidence status
sop_mappings	requirement -> SOP/process/version coverage, gap/conflict state
quality_events	deviation/OOS/OOT/finding/change, chronology, risk, status, linked evidence
investigation_records	hypotheses, evidence/gaps, reviewer disposition; AI contribution refs
capa_actions	root-cause candidate refs, action, owner, due, implementation, effectiveness evidence, closure
training_impacts	change/source version, affected role, training requirement, completion evidence
ai_uses	model/version, intended/excluded use, owner, dependencies, reviewer, release state
ai_runs	prompt/config/tools/input/retrieval/output/model identity, evidence refs, human use
evaluation_runs	frozen case set/version, expected/actual, grounding/bias/edge outcomes, failure location
dependency_edges	source/SOP/process/product/batch/training/AI/output/decision links + stale/blast-radius state
human_decisions	exact revision, action, rationale, residual risk, approver, timestamp
release_passports	AI use + context/version/evidence/tests/limitations/approver/state/history
audit_events	ordered cross-stream event chain with correlation/revision IDs
exports	inspection/evidence/forensics/passport manifest + redactions + release flag

API surface - proposed

Method / route	Purpose	Safety / state rule
POST /sources	Register controlled source/version.	Tenant/permission/type/purpose gate.
POST /requirements/extract	Extract structured requirements with source spans.	No accepted requirement without evidence.
POST /mappings	Map requirement to SOP/process/version.	Missing rationale -> review.

11A. Data model / database schema - MDM readiness

This rule extends the existing Core data objects without replacing LEDGER-Q local identifiers, release/passport semantics or tenant controls.

Entity	MDM-ready schema rule	Cross-module purpose
Product	local Product_ID + optional external_uuid + external_system + mapping_status	Correlate product quality, batch, CAPA and AI-assurance context with a future enterprise master.
Site	local Site_ID + optional external_uuid + external_system + mapping_status	Correlate facility/site context across quality, regulatory and manufacturing modules.
User	local User_ID + optional external_uuid + external_system + mapping_status	Support future enterprise identity/master-data correlation while keeping authorization/approval authority separate.

Mandatory MDM rule

All local entities (Product_ID, Site_ID, User_ID) must be designed to accept external UUIDs for future Master Data Management (MDM) cross-module orchestration readiness.

- External UUIDs are optional correlation attributes; local IDs and historical evidence remain stable.
- Store mapping source/system, mapping status, effective time and mapping history. Suggested states: UNMAPPED, MATCHED, CONFLICT, RETIRED.
- An external identity match does not grant data access, QA authority or signature rights; tenant/role/purpose controls still apply.
- Cross-module dependency/blast-radius logic may consume the external UUID only after an authorized mapping is established.

Architecture outcome LEDGER-Q is prepared for future MDM-based orchestration across PRAMANEX modules without pretending to be the enterprise master-data authority itself.

Method / route	Purpose	Safety / state rule
POST /events	Create deviation/OOS/OOT/finding/change.	Typed event + chronology; no AI root-cause decision.
POST /impact/calculate	Compute bidirectional dependency/blast radius.	Marks stale/reopens; never silently rewrites old approvals.
POST /capa	Create/update CAPA action/effectiveness.	Human closure + evidence required.
POST /ai-uses	Register intended use/model/context/release state.	No VALID state without required evidence/review.
POST /ai-runs	Record AI provenance/contribution.	Immutable run identity + exact evidence.
POST /evaluations/run	Execute frozen/adverse evaluation set.	Denominators/failures preserved.
POST /reviews/{id}/decision	Human QA approve/revise/request/reject/suspend/escalate.	Expected-revision guard.
POST /revalidate	Run affected critical retest after change.	Release state updates only after review.
POST /exports	Generate inspection/forensics/passport package.	Approved, permission-filtered, version-bound evidence only.

Bounded agent / specialist model

Specialist	Job	Must not do
Requirement mapping specialist	Extract/map requirement evidence and gaps.	Approve compliance or invent coverage.
Deviation/OOS specialist	Organize chronology, evidence, missing data and hypothesis candidates.	Determine root cause/disposition.
Change-impact specialist	Traverse dependency graph and propose blast radius/stale objects.	Automatically close or release affected controls.
CAPA/effectiveness specialist	Organize action/evidence and effectiveness candidate checks.	Autonomously approve CAPA closure.
AI provenance specialist	Record/reconstruct model/prompt/tools/data/retrieval/output/human use.	Rewrite prior runs or hide configuration.
AI evaluation specialist	Run fixed grounding/hallucination/bias/edge/regression assessments.	Hide critical failure behind aggregate score.
Evidence / audit verifier	Validate source/version/evidence and reconstruction/export completeness.	Override named human QA authority.

Agentic design rule Use agents only for bounded candidate work. Deterministic checks, permissions, state transitions, persistence and release gates remain code/workflow responsibilities. No decorative swarm and no autonomous regulated disposition.

12. Automation runtime and state machine

One parent workflow owns trigger, authorization, idempotency, parsing, specialist routing, validation, persistence, human waits, retries, export and terminal state. Every work item has a correlation/run ID and exact revision.

- START -> AUTHORIZED -> INPUT_VERSIONED -> PARSED -> CANDIDATE_READY -> VALIDATED -> HUMAN_REVIEW_REQUIRED -> APPROVED/REVISED/REQUEST_EVIDENCE/REJECTED -> ACTION_OPEN -> VERIFIED/CLOSED or HOLD/EXHAUSTED.
- Retries are bounded; an exhausted path is visible and owned. Never convert an error into a success state.
- Revision changes invalidate stale approvals. A human decision is bound to an exact source/input/output revision.
- Notifications use an outbox/idempotency key. A retry must not create duplicate external effects.

13. Validation, tests and measurable outcomes

Test family	Required cases	Pass meaning
Source / requirement	authorized source; wrong purpose; exact span; stale version	Only permitted evidence enters; requirement trace is exact.
Mapping / gap	covered requirement; missing SOP; conflicting versions	Gap/conflict remains visible; no auto-compliance.
Deviation / OOS	complete chronology; missing evidence; AI-supported hypothesis	AI contribution reconstructable; root cause remains human.
Change / blast radius	SOP change -> AI stale; model change -> training/SOP review	Both directions reopen correct dependencies only.
CAPA / effectiveness	implemented action; no effectiveness evidence; recurrence	Closure blocked until human evidence-backed decision.
Human correction firewall	correct model output; attempt silent memory/benchmark update	Correction becomes governed evidence, not automatic training.
AI grounding	supported answer; missing citation; unsupported claim	Unsupported output flagged/rejected.
Bias / edge / consistency	subgroup cases; repeated runs; model version comparison	Denominators and critical failures visible.
Release passport	valid -> source/model change -> suspended/expired -> retest -> revalidated	State transitions bound to evidence/version/human approval.
Assurance debt	missing/stale/conflicting evidence	Debt visible with exact affected dependencies.
Tenant/role	wrong tenant/role retrieval/review/export	No protected evidence leakage.
Restore/replay	restore and reconstruct quality+AI decision	Original source/run/decision/passport identifiers match.

Suggested measurable outcomes for the prototype

Metric	Definition / denominator
Requirement trace coverage	accepted requirements with exact source/version/location / accepted requirements.
Open assurance debt	count and age of missing/stale/conflicting evidence items, by severity.
Quality event review completeness	events with required chronology/evidence/reviewer fields / events reviewed.
CAPA effectiveness completion	CAPAs with required effectiveness evidence and human closure / CAPAs due.
Reopened dependency precision	correctly reopened objects / all reopened objects in frozen change cases.
AI grounding pass rate	supported/cited expected cases passed / total grounding cases; list critical failures separately.
Unsupported output escape rate	unsupported outputs incorrectly accepted / unsupported challenge cases.
Abstention / request-evidence rate	abstained or evidence-request cases / evaluation cases; never report without task success.
Critical edge-case pass	critical cases passed / critical cases; cannot be hidden by aggregate score.
Release-state freshness	VALID AI uses with current required dependencies/tests / all VALID AI uses.
Human override/correction rate	reviewer overrides/corrections / reviewed AI-assisted outputs; show

Metric	Definition / denominator
	rationale categories.
Inspection reconstruction completeness	required source/event/action/decision evidence recovered / required reconstruction items.

Claim discipline Synthetic/public-safe results prove only the tested fixture behavior. They do not prove production validation, real-world ROI, regulator acceptance, customer adoption or enterprise security.

14. Security, privacy, recovery and enterprise hardening

- Least privilege: role and tenant checks before retrieval, review, export or model access.
- Network intake: allow-listed destinations; redirect recheck; private-network block; timeout and payload-size controls.
- Concurrency: expected-revision compare/update for all approvals and corrections.
- Secrets: server-side secret manager/environment; never expose API keys in browser bundles, repositories, screenshots or exports.
- Audit: append ordered event history, but do not claim tamper-proof without a defined threat model and tested external controls.
- Backups: restore test must reproduce at least one historical decision/package using original identifiers and hashes.
- Privacy: public demo uses labelled synthetic/public-safe data. Confidential company data needs contractual, security and environment controls outside the portfolio demo.
- Accessibility: keyboard navigation, focus states, semantic labels, contrast, reduced-motion support and non-color status cues.

15. MVP -> controlled live demo -> enterprise path

Gate	Required evidence
Specified	Feature/state/data/API/acceptance behavior documented.
Built	Runnable component + setup instructions + actual artifact.
Tested	Executed expected-vs-actual tests; failures and limitations retained.
MVP	One useful bounded end-to-end journey works with synthetic/public-safe data and named human review.
Controlled live demo	Hosted endpoint opened and checked; worker/scheduler works without browser; access rules and operating limits recorded.
Pilot	Real organization accepts intended use, security, recovery, support, privacy and success criteria.
Production	Validated operating controls appropriate to intended regulated use. Not implied by portfolio completion.

Quality target: do not promise a permanently bug-free system. Before demo release, require zero known P0/P1 defects in the tested critical journey, no uncaught errors on tested routes, passing tenant/role negatives, tested failure states and reproducible test evidence.

16. Gemini master build prompt - paste-ready

MASTER BUILD TASK - PRAMANEX LEDGER-Q

Build a working controlled SaaS prototype for LEDGER-Q - Quality, Compliance & AI Assurance OS. Preserve canonical F01-F20 and shared L01-L12. Implement LQ-P01-LQ-P05 and LQ-R01-LQ-R04 only as clearly labeled child/proposed controls; do not create extra top-level features/layers.

DUAL CORE

QUALITY: source -> requirement -> gap/event -> investigation -> risk -> RCA/CAPA/change -> training/action -> human QA decision -> effectiveness -> inspection evidence.

AI ASSURANCE: AI use/model -> intended use/context -> provenance -> evaluation/grounding -> bias/edge/consistency -> risk -> human override -> release state -> monitor/change -> retest.

CONVERGENCE: any material quality or AI change traverses the evidence/dependency graph, marks affected assurance stale, reopens impacted controls, and requires human-reviewed retest before updated release/export.

NON-NEGOTIABLES

- Human QA authority remains final for regulated disposition, root cause, closure, suspension/release and residual-risk acceptance.
- Preserve exact source/version evidence and exact AI run provenance.
- Missing/conflicting/stale evidence becomes assurance debt; UNKNOWN is never scored as safe.
- Human correction cannot silently self-train, rewrite prompts/policy/knowledge or alter benchmarks. Reuse only through controlled verify -> classify -> approve -> version -> retest.
- Critical paired failures cannot be hidden by aggregate metrics. Report denominators, abstention and failure cost.
- Historical decisions/runs remain reconstructable using the versions that existed at the time.
- Basic training-impact traceability F10 is in scope; do not build the separate standalone GxP Training AI OS now.

UI

Create a premium enterprise shell with Assurance Command Center, Controlled Sources, Requirements & SOP Map, Quality Events, Change & Blast Radius, RCA-CAPA, Training Impact, Inspection Readiness, PQR/APQR Trends, AI Use & Release Registry, AI Evaluation Lab, Human QA Review Queue, Replay/Forensics/Export, Settings. Public landing may use restrained 3D dual-stream/evidence-graph visualization; operational screens prioritize evidence, status, risk and human action.

IMPLEMENTATION

Typed FastAPI contracts; PostgreSQL/Supabase-style auth + tenant RLS; versioned object storage; evidence/dependency graph represented in relational edges (vector search optional, never authoritative); n8n parent workflow with bounded named specialists; fixed/adverse test packs; one controlled write path; full correlation/revision IDs.

BUILD ORDER

1 inspect repo; 2 schema + state machine; 3 source/requirement registry; 4 mapping/gap controls; 5 deviation/OOS event flow; 6 risk/change/blast-radius; 7 CAPA/training/action; 8 human QA review; 9 AI registry/provenance; 10 evaluation/grounding/bias/edge; 11 release passport + revalidation; 12 audit/forensics/export; 13 UI; 14 adverse/security/recovery tests; 15 deploy controlled demo + verify endpoint.

USAGE OPTIMIZATION

Patch rather than rewrite; reuse schemas/components/workflows; deterministic rules before model calls; cache by source/model/config revision; retrieve minimum evidence needed; avoid decorative agents and verbose build commentary. Return concise PLAN / CHANGES / TESTS / STATUS.

SUCCESS

A user can trace a controlled requirement into quality work, run a quality event through evidence/risk/CAPA/human QA/effectiveness, register and evaluate an AI use, see bidirectional change invalidation/blast radius, preserve human corrections without self-training, update a version-bound release state only after retest/review, and reconstruct/export the complete evidence chain with no known P0/P1 defects in the tested critical journey.

17. Final acceptance checklist

- Canonical feature IDs and meanings preserved.
- 12 shared layers reused; no unnecessary parallel architecture.
- Proposed enhancements clearly labeled as proposed until built/tested.
- Every material output preserves exact evidence/source/version linkage.
- Human authority is explicit at every regulated decision/closure/release boundary.
- Normal path and adverse path both demonstrated.
- Stale revision, missing evidence, conflict, wrong tenant/role and retry exhaustion are visible states.
- Website journeys match backend states; no fake buttons or decorative workflows in the controlled demo.
- Frontend, API, database, orchestration and workers share typed contracts and correlation IDs.
- Test evidence, screenshots, README/case study, limitations and deployment status are truthful.